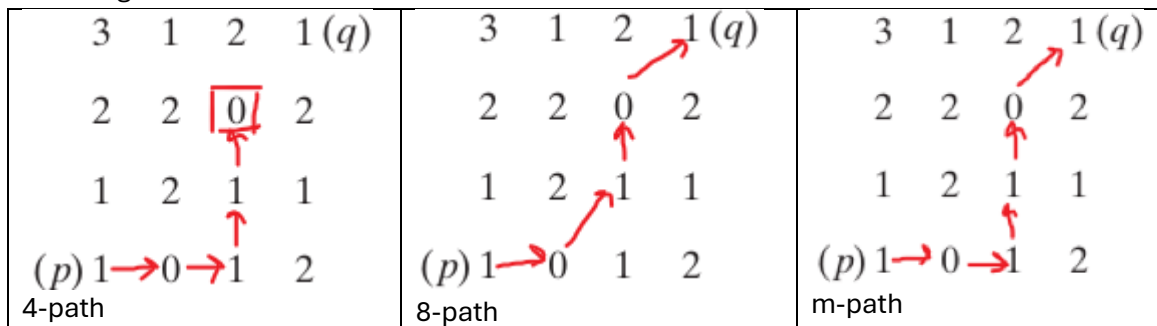
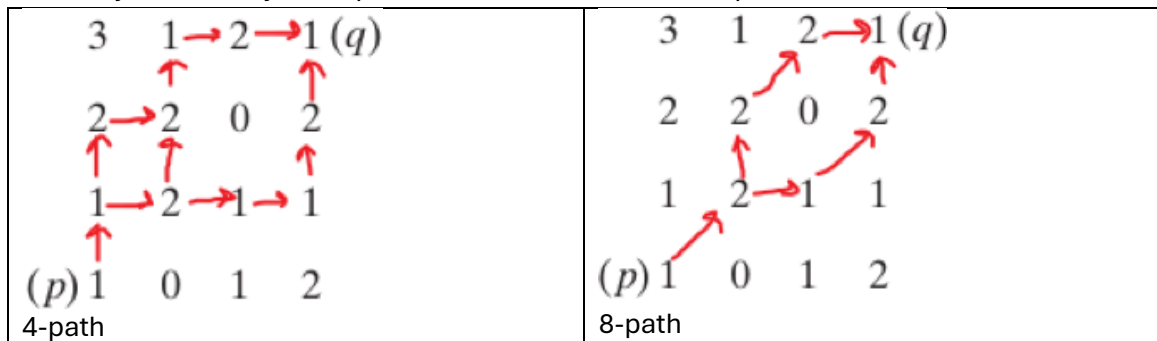


4TN4 Quiz 1

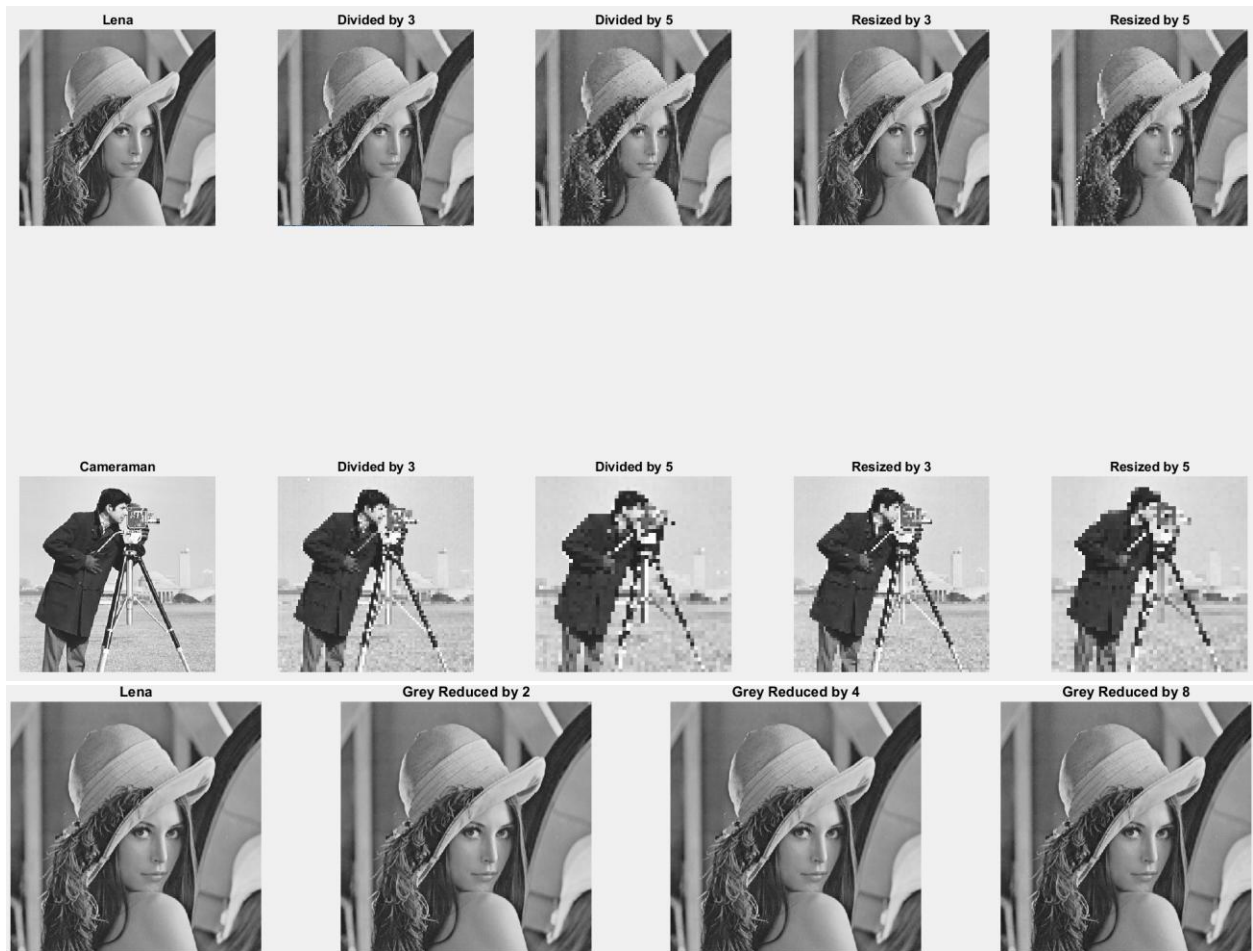
- The equation for images is $f(x, y) = i(x, y)r(x, y)$. In this case, $r(x, y) = R = 1.0$. $\therefore f(x, y) = i(x, y) = 255 * e^{-(x-x_0)^2 + (y-y_0)^2}$. Quantizing happens w/ 2^k bits. We know that our increment of intensity $\Delta I = 8$ (before the contours become discernable).
Since our interval is $[0, L-1]$, and our maximum value (K) is 255, we know that L must be 256. $\therefore$ we know that the difference in the values will be discernable when the difference in intensity is less than $\frac{L}{2^k}$, or in other words, $\Delta I \leq \frac{L}{2^k}$. So, we know that $2^k \leq L/\Delta I$, or $k \leq \frac{\log(\frac{L}{\Delta I})}{\log(2)} = \frac{\log(\frac{256}{8})}{\log(2)} = 5$.
Therefore, at a k value of 5 or below there would be visible contouring.
- The algorithm can follow the simple steps:
 - Moving diagonal UP and RIGHT $\rightarrow$ move UP and then RIGHT, or RIGHT and then UP
 - Moving diagonal UP and LEFT $\rightarrow$ move UP and then LEFT, or LEFT and then UP
 - Moving diagonal DOWN and RIGHT $\rightarrow$ move DOWN and then RIGHT, or RIGHT and then DOWN
 - Moving diagonal DOWN and LEFT $\rightarrow$ move DOWN and then LEFT, or LEFT and then DOWN
 The other movements are already legal for a 4-path algorithm, so they do not need to be changed.
- There is no 4-path between (p) and (q), since there are no paths that have points that are 4-neighbours and in the set $V\{0,1\}$. There is an 8-path that is 4 units long. There is an m-path that is 5 units long.



- There are 4 4-paths between (p) and (q), all of which take 6 units in $V\{1,2\}$. There are 2 8-paths which are 4 units long. There is an m-adjacent path, since one of the conditions for m-adjacency is if that each point "s" is in $N_4(t)$, for each point s and t on the path. Since that is already satisfied by the 4-path condition, we have an m-path.



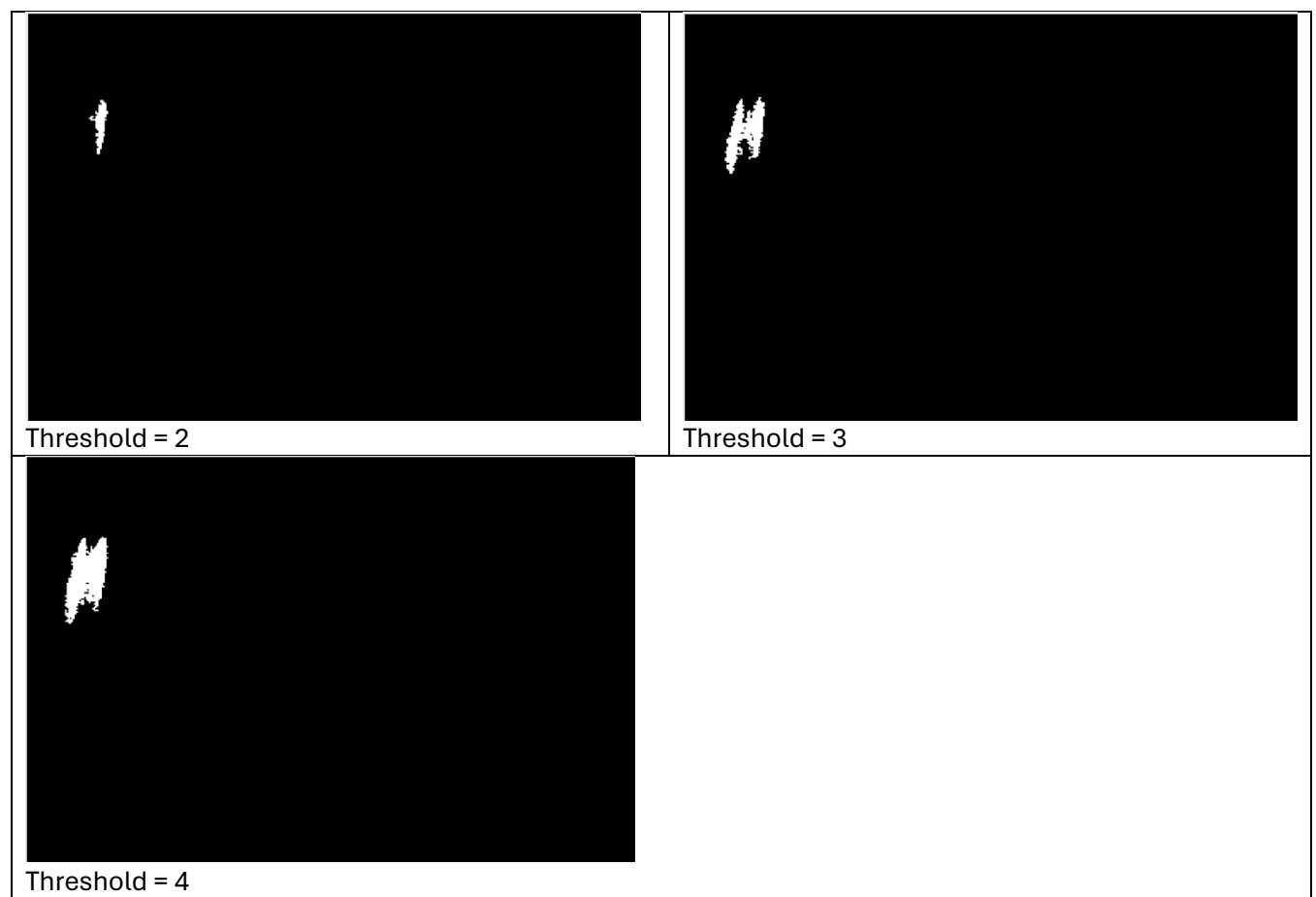
4. a) The condition where the D4 distance is equal to the shortest 4-path distance would be when we have a path that changes direction at most once. This would mean that that path would follow a straight line for a number of points, and turn either up, down, left, or right (excluding the original direction) and continue for a number of points until arriving at q. This would have to mean that all other 4-paths would not be legal
- b) This path would not necessarily be unique, since for example if there were 2 equidistant paths, both going from the bottom left corner to the top right corner, then the paths would not be unique but still satisfy the condition. In other words, if in a 5x5 square, we could traverse 5 units up followed by 5 units right, or 5 units right then 5 units up
- 5.



6.



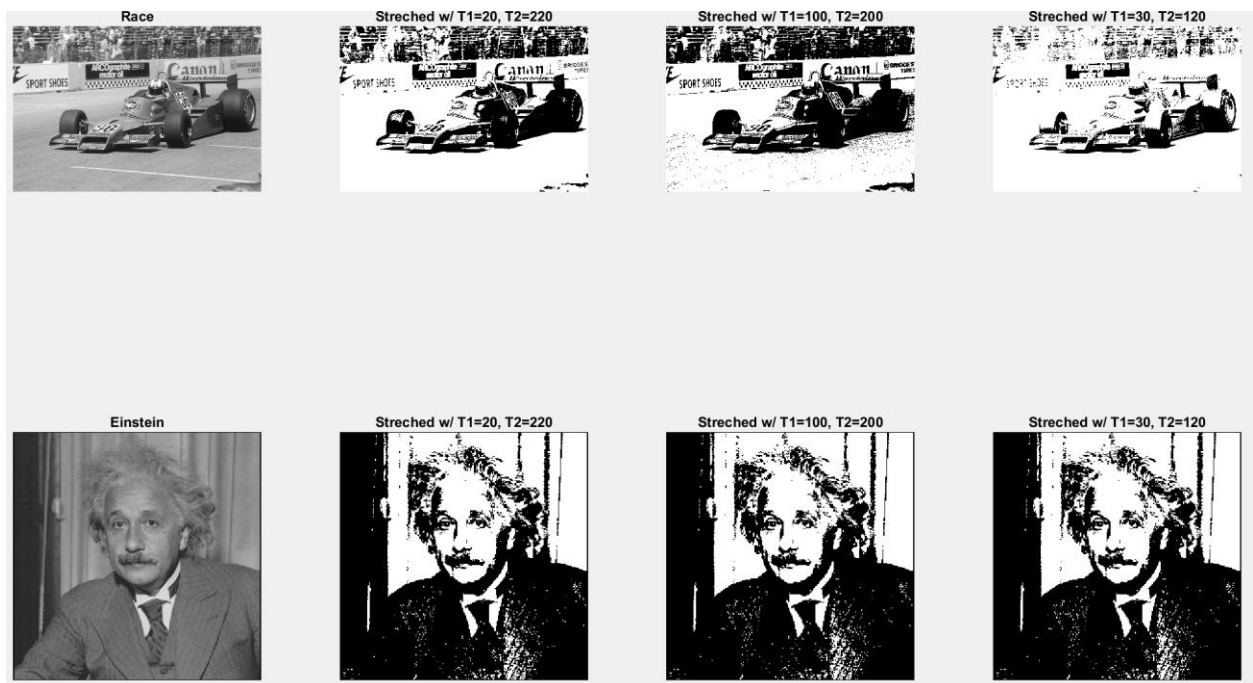
7.



8. DNC

9. DNC

10.



11.

